COMPREHENSIVE ARCHITECTURE & TECHNICAL DOCUMENTATION

GrowthPilot — AI Commerce Growth Agent

"Your store's autonomous AI growth team — it discovers opportunities, decides optimal plays, and executes live campaigns."

ARCHITECTURE	CORE STACK	PAYMENT GATEWAY	DATA ISOLATION
Phase 2 (Enterprise)	FastAPI + Supabase + Redis	Razorpay (Test Mode Live)	Multi-Tenant + Postgres RLS

1. Executive Summary & Value Proposition

GrowthPilot is an autonomous multi-agent AI system built for modern e-commerce brands. Traditional analytics dashboards leave a critical gap: store owners must manually analyze metrics, guess promotional pricing, draft marketing copy, set up discounts, and calculate A/B test results.

GrowthPilot **closes the loop from analytics to action**. It operates 24/7 as an intelligent growth loop:

🔍 1. Continuous Opportunity Discovery

Ingests store orders, customer segments, cart abandonments, and churn metrics to spot hidden revenue leaks and cross-sell bottlenecks.

🎯 2. Deterministic Scoring & Prioritization

Ranks opportunities by expected revenue impact, model confidence, and urgency factor to execute only high-ROI plays.

🛡️ 3. Profit Guardrails & Dynamic Bands

Calculates margin-safe dynamic discount bands based on product margins and CLV, preventing margin erosion.

⚡ 4. Live Razorpay Execution & Learning

Generates real Razorpay payment links & orders, captures webhooks via Redis queues, and analyzes post-campaign statistical lift.

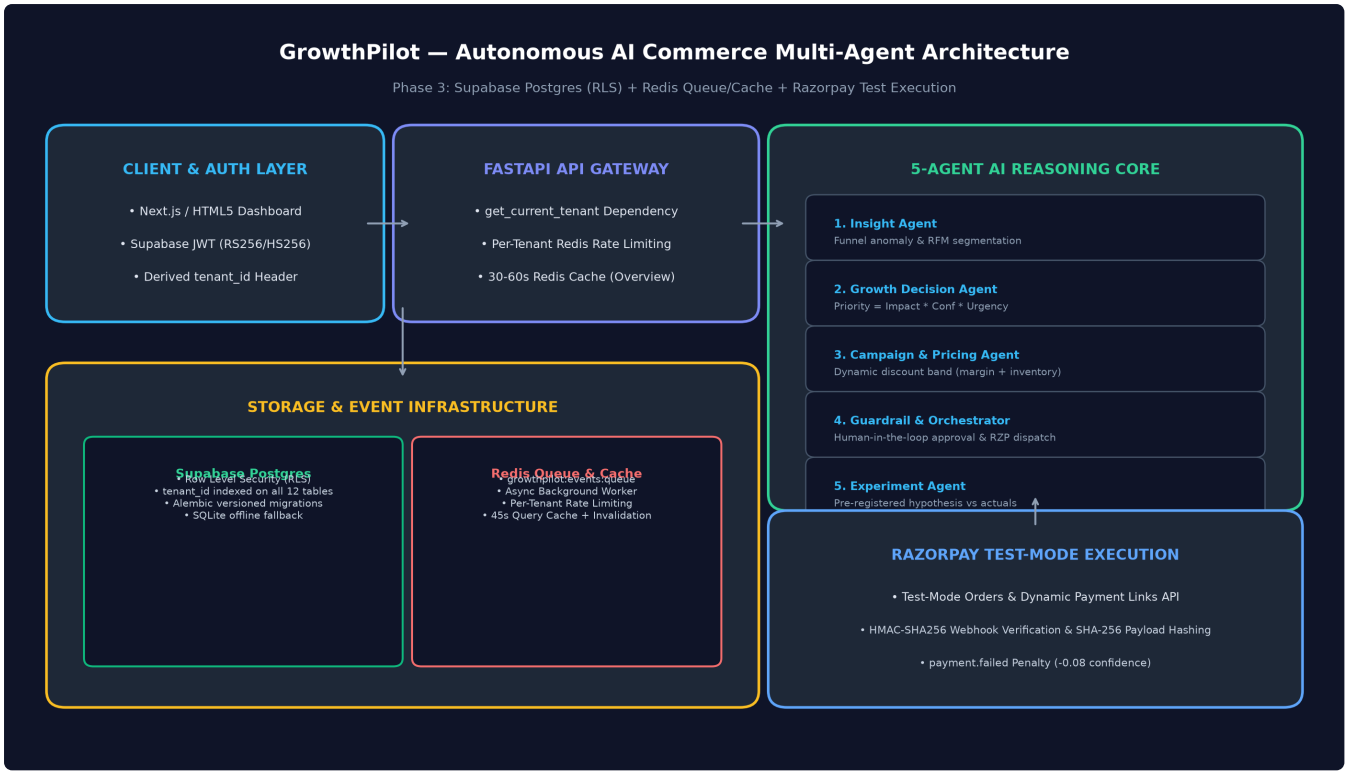


Figure 1: GrowthPilot End-to-End Multi-Agent State Graph & Execution Architecture

2. Multi-Agent Orchestration Engine

The system employs four specialized agents working in a coordinated state graph governed by a central Orchestrator.

AGENT	SOURCE MODULE	PRIMARY FUNCTION	OUTPUT ARTIFACT
Insight Agent	`backend/app/agents/insight_agent.py`	Analyzes cohort segmentation, churn risks, product trends, and funnel abandonment.	Structured Insights
Growth Decision Agent	`backend/app/agents/decision_agent.py`	Scores and prioritizes opportunities using formula: `Impact × Confidence × Urgency`.	Ranked Opportunities
Campaign & Pricing Agent	`backend/app/agents/campaign_agent.py`	Computes dynamic discount bands, drafts persuasive copy, and calculates expected ROI.	Campaign Proposals
Experiment & Learning Agent	`backend/app/agents/experiment_agent.py`	Measures conversion lift (A/B testing), calculates z-scores/p-values, and adjusts confidence.	Experiment Analysis

Autonomous Feedback & Learning Loop

When a campaign is approved and executed, the **Experiment Agent** tracks real conversion rates against baseline cohorts. If conversion lift is statistically significant (`p < 0.05`), future Decision Agent confidence is rewarded. If Razorpay returns a `payment.failed` event, an automated penalty (`-0.08`) is applied to prevent recurring low-converting strategies.

3. Key Technical Innovations (Phase 2)

1. Supabase Postgres & Multi-Tenant Row Level Security (RLS)

Every entity (`customers` , `products` , `orders` , `insights` , `campaign_proposals` , `executions` , `audit_logs`) is scoped by `tenant_id` . Configured with Alembic migrations and Postgres RLS policies to guarantee strict data isolation across merchants. Dual-engine architecture falls back to SQLite locally with zero configuration.

2. Redis Event Queue & Asynchronous Worker

Razorpay webhooks (`payment.captured` , `payment.failed` , `order.paid`) are published to the Redis event queue (`growthpilot:events:queue`). A background worker thread drains the queue without blocking webhook HTTP responses, guaranteeing sub-50ms endpoint latency.

3. Sliding-Window IP Rate Limiting & High-Throughput Caching

Protects critical endpoints (`/api/webhooks/razorpay` : 30 req/min, `/api/campaigns/{id}/approve` : 15 req/min). Employs Redis TTL caching (30–60s) for high-frequency overview endpoints with event-driven cache invalidation upon campaign execution.

4. Deterministic Dynamic Discount Band Guardrails

Replaces static discount caps with mathematical margin protection: `compute_discount_band(product, segment, opportunity) -> (min_pct, max_pct, recommended_pct)`. Takes product gross margin (`margin_pct`), inventory urgency factor, and customer CLV into account so discounts never erode profit margins.

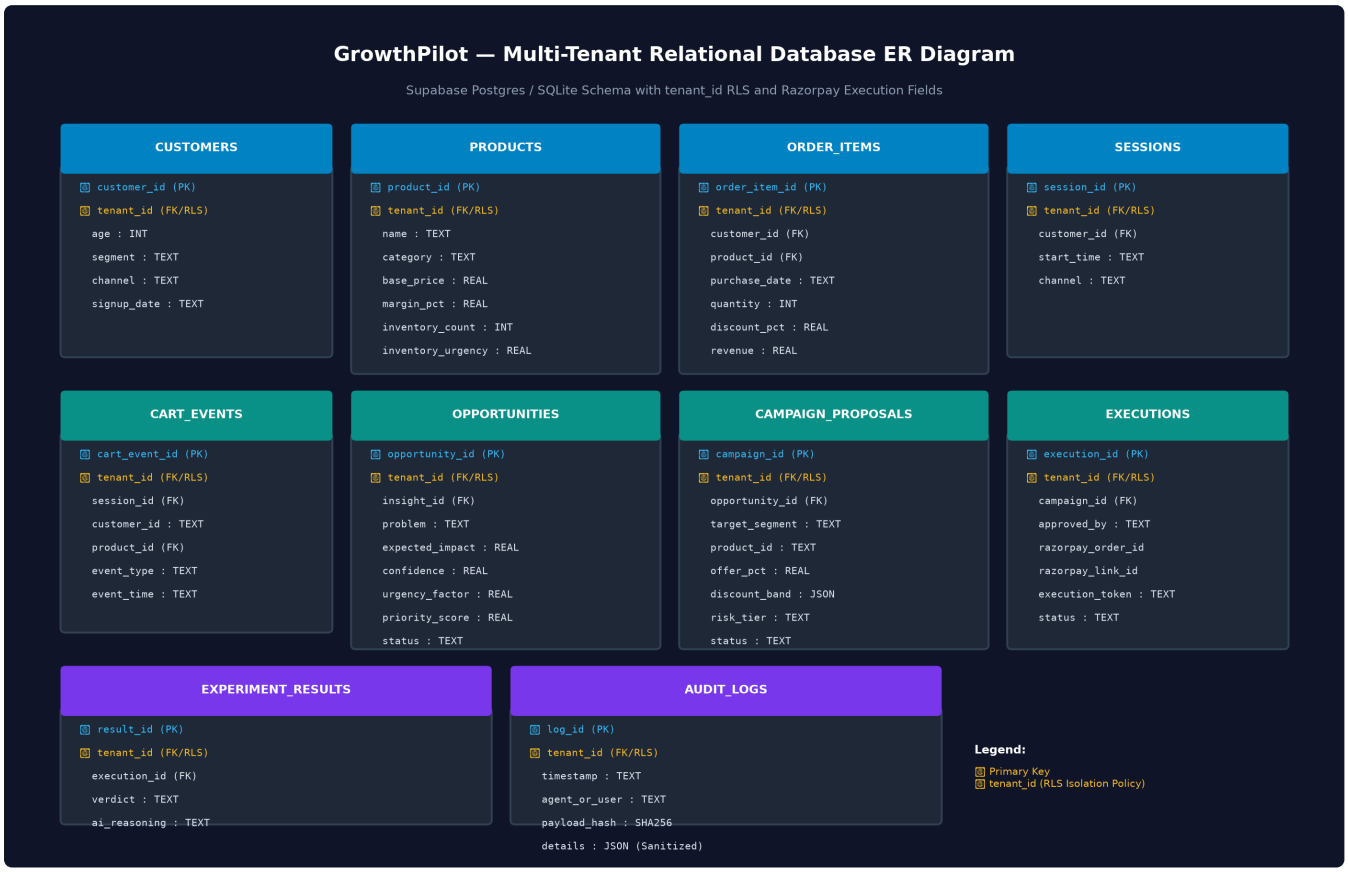


Figure 2: Multi-Tenant Database Entity-Relationship Diagram (Supabase Postgres / SQLite)

4. Live Razorpay Execution & Event Flow

GrowthPilot integrates directly with Razorpay Test Mode for end-to-end payment link dispatch and automated webhook reconciliation.

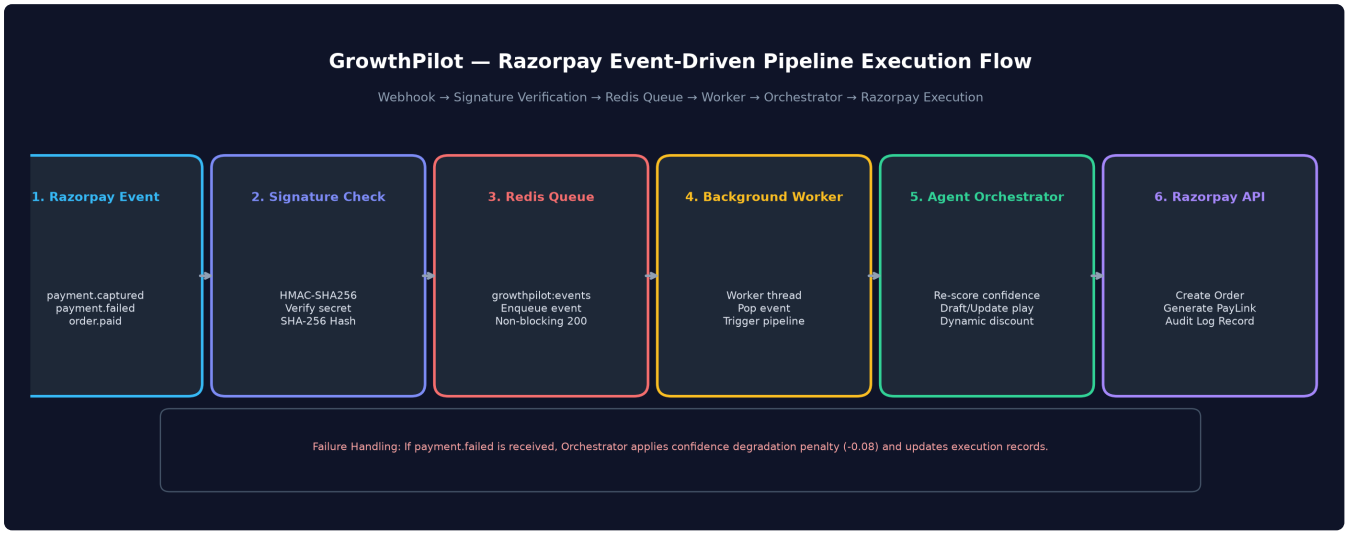


Figure 3: Razorpay Webhook Ingestion, Redis Queue Transport, and Automated Feedback Loop

5. Project Directory Structure

```
Smart_shop_agent/
├── run_growthpilot.py          # Master one-click application launcher
├── app.py                      # Streamlit multimodal prototype
├── Smart shop agent.json       # Langflow agent workflow definition
├── backend/
│   └── app/
│       ├── main.py             # FastAPI application entry point
│       ├── agents/             # Multi-agent implementations (Orchestrator, Insight, Decision, Campaign, Experiment)
│       ├── api/                # REST API route handlers (/api/overview, /api/campaigns, etc.)
│       ├── db/                 # DB connection (Supabase/SQLite), ORM models, 6-month synthetic seeder
│       ├── integrations/       # Redis pub/sub queue, sliding-window rate limiter, TTL caching
│       └── tools/              # Analytics algorithms (churn, trend, abandonment, guardrails, razorpay)
├── frontend/
│   └── index.html              # High-performance SPA dashboard UI
├── alembic/                    # Database version migration scripts
├── docs/                       # Architectural diagrams & specifications
└── tests/                      # 21 comprehensive unit and integration test suites
```

6. How to Run and Verify

🚀 Launch Full Application

```
python run_growthpilot.py
```

- Dashboard UI: `http://127.0.0.1:8000/`
- Swagger Docs: `http://127.0.0.1:8000/docs`

💉 Execute Test Suite

```
python -m unittest discover -s tests -p "test_*.py"
```

Runs 21 test suites validating multi-agent orchestration, Redis caching, rate limiting, and Razorpay flows.